CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Monday, 23 February 2026, 3:28 PM
State	Finished
Completed on	Monday, 23 February 2026, 3:29 PM
Time taken	1 min 45 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

The code given below contains instructions to print the text "**I love Apples**" to the console.

The \n in the text "I love Apples**\n**" ensures that the line breaks after printing the text "I love Apples" (which means that nothing else is printed on the same line).

Follow the steps given below to change the text, execute **compile** command and finally **execute** the file :

1. In the code given below, change the text to print "**I love Mangoes**" instead of "**I love Apples**".

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>

int main()
{
    printf("I love Mangoes\n");
    return 0;
}
```

	Expected	Got	
✔	I love Mangoes	I love Mangoes	✔

Passed all tests! ✔

Question **2**
Correct
Marked out of 1.00
Flag question

Given below is a simple program written in **C** language.

Change the text in the code given below to make the program print "**Hello C**" instead of "**Hello B**".

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>

int main()
{
    printf("Hello C");
    return 0;
}
```

	Expected	Got	
✔	Hello C	Hello C	✔

CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Monday, 23 February 2026, 3:47 PM
State	Finished
Completed on	Monday, 23 February 2026, 4:04 PM
Time taken	17 mins 7 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

The code given below contains text that prints "**DennisRitchieBrianKernighan**".

Make the suggested changes to the code so that it prints "**DennisRitchieBrianKernighan**" as shown below.

```
Dennis Ritchie
Brian Kernighan
```

To make the required changes, follow the steps given below to introduce the SPACE character and the \n new line character appropriately:

1. Insert a space between "Dennis" and "Ritchie". Make sure that no extra space or any other character apart from space are inserted.
2. Insert a \n between "Ritchie" and "Brian" Make sure that no extra space or any other character apart from \n are inserted.
3. Insert a space between "Brian" and "Kernighan". Make sure that no extra space or any other character apart from space are inserted.

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>

int main()
{
    printf("Dennis Ritchie\nBrian Kernighan");
    return 0;
}
```

	Expected	Got	
✔	Dennis Ritchie Brian Kernighan	Dennis Ritchie Brian Kernighan	✔

Passed all tests! ✔

Question **2**

As mentioned earlier, a computer program is a collection of instructions or statements.

```
}//end of the main() function - this is an example of a end-of-line comment
```

Read the code given below to understand the different types of comments. Retype in the space provided.

Given below are 3 important points regarding comments:

1. There **should not** be any space between the two forward slashes in //, i.e., / / is incorrect. Similarly, there should not be any space between the **slash** and **star** characters in /* and */, i.e., / * and * / are incorrect.
2. **Comments do not nest**, i.e., /* and */ comment has no special meaning inside a // comment.Similarly, a // comment has no special meaning inside a /* comment.
3. One should not write comments inside character literals (i.e., characters enclosed between single-quotes). Comments inside String literals (i.e., text enclosed between double-quotes) are treated as part of the String's content.

Content to be reproduced

```
/*
 This is a sample C program
 developed by REC
*/
#include <stdio.h>
int main()
{
  // this is an end of line comment
  printf("I love C Language!");
  return 0;
}
```

Answer: (penalty regime: 0 %)

```
/*
This is a sample C program
developed by REC
*/
#include <stdio.h>
int main()
{
    //this is an end of line comment
    printf("I love C Language!");
    return 0;
}
```

	Expected	Got	
✔	I love C Language!	I love C Language!	✔

Passed all tests! ✔

Question **3**
Correct
Marked out of 1.00
Flag question

In **C**, the backslash character \ is used to mark an escape sequence. An **Escape Sequence** is an escape character \ followed by a normal character. For example: \n or \t.

The presence of the escape character changes the meaning of the character which follows it. For example, when the string literal "Hello\tWorld" is printed, the result is seen as

```
Hello    World
```

In the string literal "Hello**\t**World", \t represents the **TAB** character.

Similarly, if we want to print a **double quote** inside a double-quoted string literal, we need to escape the **double quote** by using the escape character \. For example :

```
printf("Hello \" (Quote)");
```

The code given above will produce the following output:

```
Hello " (Quote)
```

Given below are a few points regarding **escape sequences**:

- Each escape sequence has a unique ASCII value as shown in the table given below.
- Each and every combination of an escape sequence starts with backslash \.
- Although an escape sequence consists of two characters, it represents a single special character in the given context.

Escape sequences and their ASCII codes:

Character	Bell	Backspace	Horizontal tab	Vertical tab	New line	Form feed	Carriage return	Double Quotation	Single Quotation	Question mark	Backslash	Null
Escape Sequence	\a	\b	\t	\v	\n	\f	\r	\"	\'	\?	\	\0
ASCII value	007	008	009	011	010	012	013	034	039	063	092	000

Escape sequences and their ASCII codes.

Character	Bell	Backspace	Horizontal tab	Vertical tab	New line	Form feed	Carriage return	Double Quotation	Single Quotation	Question mark	Backslash	Null
Escape Sequence	\a	\b	\t	\v	\n	\f	\r	\"	\'	\?	\	\0
ASCII value	007	008	009	011	010	012	013	034	039	063	092	000

Read the code given below and retype in the space provided. **Note** the effects of \t and \n in the resulting output when executed successfully.

Content to be reproduced

```
#include <stdio.h>
int main()
{
   printf("One Two");
   printf("Three\n");
   printf("Four\nFive\n");
   return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    printf("One Two");
    printf("Three\n");
    printf("Four\nFive\n");
    return 0;
}
```

	Expected	Got	
✔	One TwoThree Four Five	One TwoThree Four Five	✔

Passed all tests! ✔

Question **4**
Correct
Marked out of 1.00
Flag question

Make the following changes in the code given below:

1. Comment the statement which prints **'Mango'**.
2. Remove the comment on the statement which prints '**Banana**'.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    printf("Orange\n");
    //printf("Mango\n");
    printf("Banana\n");
    return 0;
}
```

	Expected	Got	

CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Wednesday, 25 February 2026, 8:07 AM
State	Finished
Completed on	Wednesday, 25 February 2026, 8:13 AM
Time taken	5 mins 22 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

Read the code given below to learn naming conventions in identifiers.

For example, consider the program given below:

```
#include <stdio.h>

int main()
{
        int age = 2; // age is an integer variable

        int firstNumber = 2; // firstNumber is an integer variable

        // If there are two or more words in an identifier/variable - User can also use "camel

        int second_number = 3; // second_number is an integer variable

        // Any space cannot be used between two words of an identifier/variable; User can use u

        int _i_am_also_a_valid_identifier = 4; // _i_am_also_a_valid_identifier is an integer v

        // An identifier/variable name must be start with an alphabet or underscore (_) only, n

        printf("age = %d\n", age);
        printf("firstNumber = %d\n", firstNumber);
        printf("second_number = %d\n", second_number);
        printf("_i_am_also_a_valid_identifier = %d\n", _i_am_also_a_valid_identifier);
        return 0;
}
```

Fill in the missing code in the below program to print the values of the given variables.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    int age = 2;
    int firstNumber = 2;
    int second_number = 3;
    int _i_am_also_a_valid_identifier = 4;
    printf("age = %d\n",age ); // Fill in the missing code
    printf("firstNumber = %d\n",firstNumber ); // Fill in the missing code
    printf("second_number = %d\n",second_number ); // Fill in the missing cod
    printf("_i_am_also_a_valid_identifier = %d\n",_i_am_also_a_valid_identifi
    return 0;
}
```

	Expected	Got	
✔	age = 2 firstNumber = 2 second_number = 3 _i_am_also_a_valid_identifier = 4	age = 2 firstNumber = 2 second_number = 3 _i_am_also_a_valid_identifier = 4	✔

Passed all tests! ✔

Finish review

Quiz navigation

1

Started on	Wednesday, 25 February 2026, 8:18 AM
State	Finished
Completed on	Wednesday, 25 February 2026, 8:23 AM
Time taken	5 mins 45 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

Identify and correct the error in the code given below.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    printf("Hello, # is a preprocessor in C");
    return 0;
}
```

	Expected	Got	
✔	`Hello, # is a preprocessor in C`	`Hello, # is a preprocessor in C`	✔

Passed all tests! ✔

Question **2**
Correct
Marked out of 1.00
Flag question

Click on **Check** without correcting the code.

This results in many errors because the main function is not defined correctly.

Now, correct the spelling of the main function and submit the program once again.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    printf("Correct Me!");
    return 0;
}
```

	Expected	Got	
✔	`Correct Me!`	`Correct Me!`	✔

Passed all tests! ✔

```
{
    printf("Sample main() function with int as return type!");
    return 0;// 0 value indicates that the execution is successful

}
```

If the programmer does not specify any return type, the return type is by default considered as int.

The name of the main() function should always be in lowercase, i.e., if a function is written as Main(), it is not the main function which is called by the **OS**.

Read the code given below to familiarize yourself with the syntax of main() function. Retype in the space provided.

```
#include <stdio.h>

int main()
{
   printf("Impossible is nothing!");
   return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    printf("Impossible is nothing!");
    return 0;
}
```

	Expected	Got	
✔	Impossible is nothing!	Impossible is nothing!	✔

Passed all tests! ✔

Question **4**
Correct
Marked out of 1.00
Flag question

Identify and correct the error in the code given below.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    printf("Hello, float data type allocates 4 bytes in memory");
    return 0;
}
```

	Expected	Got
✔	Hello, float data type allocates 4 bytes in memory	Hello, float data type allocates 4 by

Passed all tests! ✔

Passed all tests! ✔

Question **4**
Correct
Marked out of 1.00
Flag question

Identify and correct the error in the code given below.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    printf("Hello, float data type allocates 4 bytes in memory");
    return 0;
}
```

	Expected	Got
✔	Hello, float data type allocates 4 bytes in memory	Hello, float data type allocates 4 by

Passed all tests! ✔

Question **5**
Correct
Marked out of 1.00
Flag question

Identify and correct the error in the code given below.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>
int main()
{
    printf("Hello, I am learning C Language!");
    return 0;
}
```

	Expected	Got	
✔	Hello, I am learning C Language!	Hello, I am learning C Language!	✔

Passed all tests! ✔

Finish review

Time taken 29 mins 15 secs
Name PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

Identify the error and correct the code. [**Hint**: Verify if all variables are declared before they are first used.]

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    int number1 = 20, number2 = 30;
    int sub;
    sub = number1 - number2;
    printf("The difference of the two given numbers = %d\n", sub);
    return 0;
}

```

	Expected	Got
✔	The difference of the two given numbers = -10	The difference of the two given numbers =

Passed all tests! ✔

Question **2**
Correct
Marked out of 1.00
Flag question

In the program given below, we shall learn how to assign values to int data type from binary, octal, hex and character literals.

Read the code given below and retype in the space provided.

```
#include <stdio.h>

int main()

{

   int binaryThree = 0b11;

   printf("binaryThree value = %d\n", binaryThree);

   int octalEight = 010;

   printf("octalEight value = %d\n", octalEight);

   int hexTen = 0xA;

   printf("hexTen value = %d\n", hexTen);

   int asciiValueOfOne = '1';

   printf("asciiValueOfOne value = %d\n", asciiValueOfOne);

   int asciiValueOfA = 'A';

   printf("asciiValueOfA value = %d\n", asciiValueOfA);

   return 0;

}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int binaryThree=0b11;
    printf("binaryThree value = %d\n",binaryThree);
    int octalEight=010;
    printf("octalEight value = %d\n",octalEight);
    int hexTen=0xA;
    printf("hexTen value = %d\n",hexTen);
    int asciiValueOfOne='1';
    printf("asciiValueOfOne value = %d\n",asciiValueOfOne);
    int asciiValueOfA='A';
    printf("asciiValueOfA value = %d\n",asciiValueOfA);
    return 0;
}
```

Passed all tests! ✔

Question **3**
Correct
Marked out of 1.00
Flag question

In the program given below, fill in the missing code to add two integer numbers.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    int num1 = 15, num2 = 25, sum;
    printf("Given integers are num1 = %d, num2 = %d\n", num1, num2);
    sum = num1+num2;
    printf("Sum of 2 given numbers = %d\n", sum);
    return 0;
}
```

	Expected	Got	
✔	Given integers are num1 = 15, num2 = 25 Sum of 2 given numbers = 40	Given integers are num1 = 15, num2 = 25 Sum of 2 given numbers = 40	✔

Passed all tests! ✔

Question **4**
Correct
Marked out of 1.00
Flag question

To print unsigned values on the console, use %u format character instead of %d in the **printf()** function.

Whenever an attempt is made to assign a negative number to an **unsigned int** (For eg: unsigned int num = -1;) the compiler does not flag it as an **error**. Instead, it will automatically convert the negative number to a positive number as shown below:

```
unsigned int num = -1;
The value stored in num = unsigned int maximum_value + 1 - num;
The final value in num = 4294967295 (in a 32-bit processing system)
```

In the program given below, fill in the missing **format characters** to print **signed** and **unsigned** values.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    signed int number1 = -20, number2 = 20;
    unsigned int number3 = -1, number4 = 1;
    printf("Given signed values are %d and %d\n", number1, number2);
    printf("Given unsigned values are %u and %d\n", number3, number4);
    return 0;
}
```

	Expected	Got	
✔	Given signed values are -20 and 20 Given unsigned values are 4294967295 and 1	Given signed values are -20 and 20 Given unsigned values are 4294967295 and 1	✔

Passed all tests! ✔

Question **5**

To print unsigned values on the console, use %u format character instead of %d in the **printf()** function.

To print unsigned values on the console, use %u format character instead of %d in the **printf()** function.

Whenever an attempt is made to assign a negative number to an **unsigned int** (For eg: unsigned int num = -1;) the compiler does not flag it as an **error**. Instead, it will automatically convert the negative number to a positive number as shown below:

```
unsigned int num = -1;
The value stored in num = unsigned int maximum_value + 1 - num;
The final value in num = 4294967295 (in a 32-bit processing system)
```

In the program given below, fill in the missing **format characters** to print **signed** and **unsigned** values.

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>

int main()
{
    signed int number1 = -20, number2 = 20;
    unsigned int number3 = -1, number4 = 1;
    printf("Given signed values are %d and %d\n", number1, number2); // Fill
    printf("Given unsigned values are %u and %d\n", number3, number4); // Fil
    return 0;
}
```

	Expected	Got	
✔	Given signed values are -20 and 20 Given unsigned values are 4294967295 and 1	Given signed values are -20 and 20 Given unsigned values are 4294967295 and 1	✔

Passed all tests! ✔

To print unsigned values on the console, use %u format character instead of %d in the **printf()** function.

Whenever an attempt is made to assign a negative number to an **unsigned int** (For eg: unsigned int num = -1;) the compiler does not flag it as an **error**. Instead, it will automatically convert the negative number to a positive number as shown below:

```
unsigned int num = -1;
The value stored in num = unsigned int maximum_value + 1 - num;
The final value in num = 4294967295 (in a 32-bit processing system)
```

In the program given below, fill in the missing **format characters** to print **signed** and **unsigned** values.

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>

int main()
{
    signed int number1 = -20, number2 = 20;
    unsigned int number3 = -1, number4 = 1;
    printf("Given signed values are %d and %d\n", number1, number2); // Fill
    printf("Given unsigned values are %u and %d\n", number3, number4); // Fil
    return 0;
}
```

	Expected	Got	
✔	Given signed values are -20 and 20	Given signed values are -20 and 20	✔

The final value in num = 4294967295 (in a 32-bit processing system)

In the program given below, fill in the missing **format characters** to print **signed** and **unsigned** values.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    signed int number1 = -20, number2 = 20;
    unsigned int number3 = -1, number4 = 1;
    printf("Given signed values are %d and %d\n", number1, number2); // Fill
    printf("Given unsigned values are %u and %d\n", number3, number4); // Fil
    return 0;
}
```

	Expected	Got	
✔	Given signed values are -20 and 20 Given unsigned values are 4294967295 and 1	Given signed values are -20 and 20 Given unsigned values are 4294967295 and 1	✔

Passed all tests! ✔

Question **7**
Correct
Marked out of 1.00
Flag question

To print unsigned values on the console, use %u format character instead of %d in the **printf()** function.

Whenever an attempt is made to assign a negative number to an **unsigned int** (For eg: unsigned int num = -1;) the compiler does not flag it as an **error**. Instead, it will automatically convert the negative number to a positive number as shown below:

```
unsigned int num = -1;
The value stored in num = unsigned int maximum_value + 1 - num;
The final value in num = 4294967295 (in a 32-bit processing system)
```

In the program given below, fill in the missing **format characters** to print **signed** and **unsigned** values.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    signed int number1 = -20, number2 = 20;
    unsigned int number3 = -1, number4 = 1;
    printf("Given signed values are %d and %d\n", number1, number2); // Fill
    printf("Given unsigned values are %u and %d\n", number3, number4); // Fil
    return 0;
}
```

	Expected	Got	
✔	Given signed values are -20 and 20 Given unsigned values are 4294967295 and 1	Given signed values are -20 and 20 Given unsigned values are 4294967295 and 1	✔

Passed all tests! ✔

Finish review

Started on	Wednesday, 25 February 2026, 8:57 AM
State	Finished
Completed on	Wednesday, 25 February 2026, 9:04 AM
Time taken	7 mins 15 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

Identify and correct the errors in the code given below:

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    float num1 = 5.345, num2 = 12.4, result;
    printf("Given float values are num1 = %f, num2 = %f\n", num1, num2);
    result = num1 / num2;
    printf("Result of division = %f \n", result);
    return 0;
}
```

	Expected	Got
✔	Given float values are num1 = 5.345000, num2 = 12.400000 Result of division = 0.431048	Given float values are num1 = 5 Result of division = 0.431048

Passed all tests! ✔

Question **2**
Correct
Marked out of 1.00
Flag question

Identify and correct the errors in the code given below:

Expected Output:

Given float values are num1 = 5.340000, num2 = 125.789001
The result after dividing in float format = 23.555992
The result after dividing in exponential format = 2.355599e+01

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    float num1 = 5.34, num2 = 125.789, result;
    printf("Given float values are num1 = %f, num2 = %f\n", num1, num2);
    result = num2 / num1;
    printf("The result after dividing in float format = %f\n", result );
    printf("The result after dividing in exponential format = %e\n", result )
    return 0;
}
```

	Expected	Got
✔	Given float values are num1 = 5.340000, num2 = 125.789001 The result after dividing in float format = 23.555992 The result after dividing in exponential format = 2.355599e+01	Given float values are nu The result after dividing The result after dividing

Passed all tests! ✔

CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Wednesday, 25 February 2026, 9:05 AM
State	Finished
Completed on	Wednesday, 25 February 2026, 9:25 AM
Time taken	19 mins 40 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

Arithmetic operators can be used in expressions with operands of different data types.

For example, the **addition operator** (+) can be used on **int** and **float** data types with the result being a **float** data type.

If the same is performed on **float** and **double** data types, the result will be a **double** data type, i.e., the result will always be of the **data type** with the highest memory capacity.

Consider the following examples with different combinations of arithmetic operators and data types.

Expression	Result	Description
2 + 2.5	4.500000	int + float = float
'A' + 10	75	char + int = int
5.9 + 4.67L	10.570000	float + long double = long double

Fill in the code given below with the correct missing **format characters**:

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    printf("Result1 = %f\n", (2 + 2.5));
    printf("Result2 = %d\n", ('A' + 10));
    printf("Result3 = %Lf\n", (5.9 + 4.67L));
    return 0;
}
```

	Expected	Got	
✔	Result1 = 4.500000 Result2 = 75 Result3 = 10.570000	Result1 = 4.500000 Result2 = 75 Result3 = 10.570000	✔

Passed all tests! ✔

Question **2**
Correct
Marked out of 1.00
Flag question

The table given below demonstrates the use of various **arithmetic operators** using two variables c1 and c2 of type char with values 'A' and 'D' respectively:

Expression	Result
c1	65
c1 + c2	133
c1 + c2 + 5	138
c1 + c2 + '5'	186

In the above examples, the character 'A' is substituted with its ASCII value **65** and 'D' is substituted with 68. The character '5' is substituted with its ASCII value 53. The integer value 5 is used as it is.

The following table demonstrates the usage of various **arithmetic operators** using two variables a and b of type int with values 11 and -3 respectively:

Expression	Result
a + b	8
a - b	14
a * b	-33
a / b	-3

In the program given below, type the missing code to find the **result** of applying different **arithmetic operators** on **char** data type values.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    char c1 = 'A', c2 = 'D';
    printf("c1 = %d\n", c1);
    printf("c1 + c2 = %d\n", (c1 + c2));
    printf("c1 + c2 + 5 = %d\n", (c1 + c2 + 5));
    printf("Result = %d ", (c1 + c2 + '5'));
    return 0;
}
```

	Expected	Got	
✔	c1 = 65 c1 + c2 = 133 c1 + c2 + 5 = 138 Result = 186	c1 = 65 c1 + c2 = 133 c1 + c2 + 5 = 138 Result = 186	✔

Passed all tests! ✔

Division of one integer by another integer is referred to as **integer** division. This operation always results in an integer with truncated quotient.

If a **division** operation is carried out with two **floating point numbers** or with one **floating point number** and one **integer**, the result will be a **floating point quotient**.

The table given below demonstrates the usage of various **arithmetic operators** using two variables num1 and num2 of type float with values 12.5 and 2.0 respectively:

Expression	**Result**
num1 + num2	14.500000
num1 - num2	10.500000
num1 * num2	25.000000
num1 / num2	6.250000

Note that the **remainder operator** (%) is not applicable for **floating point numbers**.

In the program given below, type the missing code to find the **result** of applying different **arithmetic operators** on **floating point numbers**.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    float num1 = 12.5, num2 = 2.0;
    printf("Result of addition = %f\n", (num1 + num2));
    printf("Result of subtraction = %f\n", (num1 - num2));
    printf("Result of multiplication = %f\n", (num1 * num2));
    printf("Result of division = %f\n", (num1 / num2));
    return 0;
}
```

	Expected	Got	
✔	Result of addition = 14.500000 Result of subtraction = 10.500000	Result of addition = 14.500000 Result of subtraction = 10.500000	✔

%	Remainder/Modulus operator for finding remainder

Arithmetic operators are applied on **numeric operands**. Thus the operands can be **integers**, **floats** or **characters** (Since a character is internally represented by its numeric code).

The **remainder operator** (%) requires that both the operands be **integers** and the second operand be **non-zero**. Similarly the **division operator** (/) requires that the second operand be **non-zero**.

The format for usage of arithmetic operator is as follows:
`operand1operatoroperand2`

According to the coding conventions in C, a single space should be provided to the left and to the right of an operator.

The table given below demonstrates the use of various **arithmetic operators** using two variables num1 and num2 of type int with values 10 and 3 respectively:

Expression	Result
num1 + num2	13
num1 - num2	7
num1 * num2	30
num1 / num2	3
num1 % num2	1

Read the code given below to understand the usage of **arithmetic operators**. Retype in the space provided.

```
#include <stdio.h>
int main()
{
    int num1 = 10, num2 = 3;
    printf("Addition Result = %d\n", (num1 + num2));
    printf("Subtraction Result = %d\n", (num1 - num2));
    printf("Multiplication Result = %d\n", (num1 * num2));
    printf("Division Result = %d\n", (num1 / num2));
    printf("Remainder = %d", (num1 % num2));
    return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int num1 = 10, num2 = 3;
    printf("Addition Result = %d\n", (num1 + num2));
    printf("Subtraction Result = %d\n", (num1 - num2));
    printf("Multiplication Result = %d\n", (num1 * num2));
    printf("Division Result = %d\n", (num1 / num2));
    printf("Remainder = %d\n", (num1 % num2));
    return 0;
}
```

	Expected	Got	
✔	Addition Result = 13 Subtraction Result = 7 Multiplication Result = 30 Division Result = 3 Remainder = 1	Addition Result = 13 Subtraction Result = 7 Multiplication Result = 30 Division Result = 3 Remainder = 1	✔

Passed all tests! ✔

Finish review

<	Checks for less-than condition
<=	Checks for less-than-or-equals condition.
==	Checks if two values are equal
!=	Checks if two values are unequal

The format for usage of **relational** and **equality operators** is as follows:

`operand1operatoroperand2`

According to the coding conventions in C, a single space should be provided to the left and to the right of the operator.

The table given below demonstrates the use of various **relational and equality operators** using variables int num1 = 7;, float num2 = 5.5;, char ch = 'w':

Expression	Interpretation	Result Value
(num1 > 5)	true	1
((num1 + num2) <= 10)	false	0
(ch == 119)	true	1
(ch != 'p')	true	1
(ch >= 10 * (num1 + num2))	false	0

Read the code given below and retype in the space provided.

```
#include <stdio.h>


int main()
{
    int num1 = 7;
    float num2 = 5.5;
    char ch = 'w';
    printf("Result1 = %d\n", (num1 > 5));
    printf("Result2 = %d\n", ((num1 + num2) <= 10));
    printf("Result3 = %d\n", (ch == 119));
    printf("Result4 = %d\n", (ch != 'p'));
    printf("Result5 = %d", (ch >= 10 * (num1 + num2)));
    return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int num1= 7;
    float num2 = 5.5;
    char ch = 'w';
    printf("Result1 = %d\n",(num1>5));
    printf("Result2 = %d\n",((num1 + num2)<=10));
    printf("Result3 = %d\n", (ch== 119));
    printf("Result4 = %d\n",(ch!='p'));
    printf("Result5 = %d\n",(ch>= 10*(num1 +num2)));
    return 0;
}
```

	Expected	Got	
✔	Result1 = 1 Result2 = 0 Result3 = 1 Result4 = 1 Result5 = 0	Result1 = 1 Result2 = 0 Result3 = 1 Result4 = 1 Result5 = 0	✔

Passed all tests! ✔

to either false or true respectively.

Note: In C, false is represented as 0 (zero) and all non-zero values can be treated as true.

Given below are the **three** logical operators in **C**:

Operator	Description	Meaning
&&	logical AND	It returns true when both conditions are true, else, it returns false
||	logical OR	It returns true if atleast one of the conditions is true
!	logical NOT	It returns true when the given expression is false and returns false when the given expression is true

According to the coding conventions in C, a single space should be provided to the left and to the right of the operator.

The below table demonstrates the use of various **relational and equality operators** using variables int num1 = 7;, float num2 = 5.5;, char ch = 'w':

Expression	Interpretation	Result Value
(num1 >= 6) && (ch == 'w')	true	1
(num2 < 11) && (num1 > 100)	false	0
(ch != 'p') || ((num1 + num2) <= 10)	true	1
!(num1 > (num2 + 1))	false	0
!(num1 <= 3)	true	1

Read the code given below and retype in the space provided.

```
#include <stdio.h>

int main()
{
   int num1 = 7;
   float num2 = 5.5;
   char ch = 'w';
   printf("Result1 = %d\n", ((num1 >= 6) && (ch == 'w')));
   printf("Result2 = %d\n", ((num2 < 11) && (num1 > 100)));
   printf("Result3 = %d\n", ((ch != 'p') || ((num1 + num2) <= 10)));
   printf("Result4 = %d\n", !(num1 > (num2 +1)));
   printf("Result5 = %d\n", !(num1 <= 3));
   return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int num1=7;
    float num2=5.5;
    char ch = 'w';
    printf("Result1 = %d\n",((num1 >= 6) && (ch=='w')));
    printf("Result2 = %d\n", ((num2 < 11) && (num1>100)));
    printf("Result3 = %d\n", ((ch !='p')|| ((num1 +num2)<=10)));
    printf("Result4 = %d\n",!(num1 > (num2 + 1)));
    printf("Result5 = %d\n",!(num1 <= 3));
    return 0;
}
```

	Expected	Got	
✔	Result1 = 1 Result2 = 0 Result3 = 1 Result4 = 0 Result5 = 1	Result1 = 1 Result2 = 0 Result3 = 1 Result4 = 0 Result5 = 1	✔

Passed all tests! ✔

CS23232-Fundamentals of D ructures Using C-2025 Batch

Started on	Thursday, 26 February 2026, 3:45 PM
State	Finished
Completed on	Thursday, 26 February 2026, 4:07 PM
Time taken	22 mins 24 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

Read the code given below to understand the working of unary operators. Retype in the space provided.

```
#include <stdio.h>

int main()
{
   int x = 16;
   printf("+x = %d\n", (+x));
   printf("-x = %d\n", (-x));
   printf("x = %d\n", x);
   printf("++x = %d\n", (++x));
   printf("x = %d\n", x);
   printf("x++ = %d\n", (x++));
   printf("x = %d\n", x);
   printf("--x = %d\n", (--x));
   printf("x = %d\n", x);
   printf("x-- = %d\n", (x--));
   printf("x = %d", x);
   return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int x =16;
    printf("+x = %d\n",(+x));
    printf("-x = %d\n", (-x));
    printf("x = %d\n", x);
    printf("++x = %d\n",(++x));
    printf("x = %d\n",x);
    printf("x++ = %d\n", (x++));
    printf("x = %d\n",(x));
    printf("--x = %d\n",(--x));
    printf("x = %d\n", x);
    printf("x-- = %d\n", (x--));
    printf("x = %d\n",x);
    return 0;
}
```

	Expected	Got	
✔	+x = 16 -x = -16 x = 16 ++x = 17 x = 17 x++ = 17 x = 18 --x = 17 x = 17 x-- = 17 x = 16	+x = 16 -x = -16 x = 16 ++x = 17 x = 17 x++ = 17 x = 18 --x = 17 x = 17 x-- = 17 x = 16	✔

Passed all tests! ✔

Question **2**
Correct
Marked out of 1.00
Flag question

Read the code given below to understand the working of **increment** and **decrement** operators. Retype in the space provided.

```
#include <stdio.h>

int main()
```

```
    return 0;
}
```

	Expected	Got	
✔	+x = 16 -x = -16 x = 16 ++x = 17 x = 17 x++ = 17 x = 18 --x = 17 x = 17 x-- = 17 x = 16	+x = 16 -x = -16 x = 16 ++x = 17 x = 17 x++ = 17 x = 18 --x = 17 x = 17 x-- = 17 x = 16	✔

Passed all tests! ✔

Question **2**

Correct

Marked out of 1.00

Flag question

Read the code given below to understand the working of **increment** and **decrement** operators. Retype in the space provided.

```
#include <stdio.h>

int main()
{
   int x = 4, y;
   y = x++;
   printf("y = %d x = %d\n", y, x);
   y = ++x;
   printf("y = %d x = %d\n", y, x);
   y = x–;
   printf("y = %d x = %d\n", y, x);
   y = –x;
   printf("y = %d x = %d\n", y, x);
   return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int x=4,y;
    y=x++;
    printf("y = %d x = %d\n", y,x);
    y=++x;
    printf("y = %d x = %d\n", y,x);
    y = x--;
    printf("y = %d x = %d\n", y,x);
    y = --x;
    printf("y = %d x = %d\n", y,x);
    return 0;
}
```

	Expected	Got	
✔	y = 4 x = 5 y = 6 x = 6 y = 6 x = 5 y = 4 x = 4	y = 4 x = 5 y = 6 x [illegible] y = 6 [illegible] y = [illegible]	✔

Passed all tests! ✔

Fundamentals of Data Structures Using C-2025 Batch

n Thursday, 26 February 2026, 4:09 PM
e Finished
n Thursday, 26 February 2026, 4:17 PM
n 7 mins 39 secs
e PAVITHRA B 2025-BME

Read the code given below to understand the usage of the assignment operator. Retype in the space provided.

```
#include <stdio.h>


int main()

{

  int x = 24, y = 39, z = 45;

  z = x + y;

  y = z - y;

  x = z - y;

  printf("x = %d y = %d z = %d", x, y, z);

  return 0;

}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main ()
{
    int x = 24 , y = 39, z = 45;
    z = x + y;
    y = z - y;
    x = z - x;
    printf("x = %d y = %d z = %d", x,y,z);
    return 0;
}
```

	Expected	Got	
✔	x = 39 y = 24 z = 63	x = 39 y = 24 z = 63	✔

Passed all tests! ✔

Read the code given below and retype in the space provided.

```
#include <stdio.h>

int main()
{
  int x = 2, y = 18, z = 12;
  x += y;
  printf("x = %d\n", x);
  y *= 2;
  printf("y = %d\n", y);
  z /= 5;
  printf("z = %d\n", z);
  x %= 7;
  printf("x = %d", x);
  return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
```

```
    int x = 24 , y = 39, z = 45;
    z = x + y;
    y = z - y;
    x = z - x;
    printf("x = %d y = %d z = %d", x,y,z);
    return 0;
}
```

	Expected	Got	
✔	x = 39 y = 24 z = 63	x = 39 y = 24 z = 63	✔

Passed all tests! ✔

Read the code given below and retype in the space provided.

#include <stdio.h>

int main()
{
 int x = 2, y = 18, z = 12;
 x += y;
 printf("x = %d\n", x);
 y *= 2;
 printf("y = %d\n", y);
 z /= 5;
 printf("z = %d\n", z);
 x %= 7;
 printf("x = %d", x);
 return 0;
}

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int x = 2, y = 18, z = 12;
    x += y;
    printf("x = %d\n",x);
    y *= 2;
    printf("y = %d\n",y);
    z /= 5;
    printf("z = %d\n",z);
    x %= 7;
    printf("x = %d",x);
    return 0;
}
```

	Expected	Got	
✔	x = 20 y = 36 z = 2 x = 6	x = 20 y = 36 z = 2 x = 6	✔

Passed all tests! ✔

Finish review

-undamentals of Da... tructures Using C-2025 Batcl

- Thursday, 26 February 2026, 4:09 PM
- Finished
- Thursday, 26 February 2026, 4:17 PM
- 7 mins 39 secs
- PAVITHRA B 2025-BME

Read the code given below to understand the usage of the assignment operator. Retype in the space provided.

```
#include <stdio.h>


int main()

{

  int x = 24, y = 39, z = 45;

  z = x + y;

  y = z - y;

  x = z - y;

  printf("x = %d y = %d z = %d", x, y, z);

  return 0;

}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main ()
{
    int x = 24 , y = 39, z = 45;
    z = x + y;
    y = z - y;
    x = z - x;
    printf("x = %d y = %d z = %d", x,y,z);
    return 0;
}
```

	Expected	Got	
✔	x = 39 y = 24 z = 63	x = 39 y = 24 z = 63	✔

Passed all tests! ✔

Read the code given below and retype in the space provided.

```
#include <stdio.h>

int main()
{
  int x = 2, y = 18, z = 12;
  x += y;
  printf("x = %d\n", x);
  y *= 2;
  printf("y = %d\n", y);
  z /= 5;
  printf("z = %d\n", z);
  x %= 7;
  printf("x = %d", x);
  return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
```

State Finished
Completed on Thursday, 26 February 2026, 4:17 PM
Time taken 7 mins 39 secs
Name PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

Read the code given below to understand the usage of the assignment operator. Retype in the space provided.

```
#include <stdio.h>


int main()
{
  int x = 24, y = 39, z = 45;
  z = x + y;
  y = z - y;
  x = z - y;
  printf("x = %d y = %d z = %d", x, y, z);
  return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main ()
{
    int x = 24 , y = 39, z = 45;
    z = x + y;
    y = z - y;
    x = z - x;
    printf("x = %d y = %d z = %d", x,y,z);
    return 0;
}
```

	Expected	Got	
✔	x = 39 y = 24 z = 63	x = 39 y = 24 z = 63	✔

Passed all tests! ✔

Question **2**
Correct
Marked out of 1.00
Flag question

Read the code given below and retype in the space provided.

```
#include <stdio.h>

int main()
{
  int x = 2, y = 18, z = 12;
  x += y;
  printf("x = %d\n", x);
  y *= 2;
  printf("y = %d\n", y);
  z /= 5;
  printf("z = %d\n", z);
  x %= 7;
  printf("x = %d", x);
  return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
```

```
    return 0;
}
```

	Expected	Got	
✔	x = 39 y = 24 z = 63	x = 39 y = 24 z = 63	✔

Passed all tests! ✔

Read the code given below and retype in the space provided.

```
#include <stdio.h>

int main()
{
   int x = 2, y = 18, z = 12;
   x += y;
   printf("x = %d\n", x);
   y *= 2;
   printf("y = %d\n", y);
   z /= 5;
   printf("z = %d\n", z);
   x %= 7;
   printf("x = %d", x);
   return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int x = 2, y = 18, z = 12;
    x += y;
    printf("x = %d\n",x);
    y *= 2;
    printf("y = %d\n",y);
    z /= 5;
    printf("z = %d\n",z);
    x %= 7;
    printf("x = %d",x);
    return 0;
}
```

	Expected	Got	
✔	x = 20 y = 36 z = 2 x = 6	x = 20 y = 36 z = 2 x = 6	✔

Passed all tests! ✔

Finish review

Saved

ime

```
    return 0;
}
```

	Expected	Got	
✔	x = 39 y = 24 z = 63	x = 39 y = 24 z = 63	✔

Passed all tests! ✔

Question **2**

Correct

Marked out of 1.00

Flag question

Read the code given below and retype in the space provided.

```
#include <stdio.h>

int main()
{
   int x = 2, y = 18, z = 12;
   x += y;
   printf("x = %d\n", x);
   y *= 2;
   printf("y = %d\n", y);
   z /= 5;
   printf("z = %d\n", z);
   x %= 7;
   printf("x = %d", x);
   return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int x = 2, y = 18, z = 12;
    x += y;
    printf("x = %d\n",x);
    y *= 2;
    printf("y = %d\n",y);
    z /= 5;
    printf("z = %d\n",z);
    x %= 7;
    printf("x = %d",x);
    return 0;
}
```

	Expected	Got	
✔	x = 20 y = 36 z = 2 x = 6	x = 20 y = 36 z = 2 x = 6	✔

Passed all tests! ✔

Fundamentals of Data Structures Using C-2025 Batch

n Sunday, 1 March 2026, 3:05 PM
e Finished
n Sunday, 1 March 2026, 3:12 PM
n 7 mins 7 secs
e PAVITHRA B 2025-BME

C language provides an operator to evaluate conditions. It is called a **conditional operator** (?:) or a **ternary operator**.

Ternary operator needs exactly **three** operands to compute the result.

The syntax for using a **ternary operator** is:

```
condition? expression1 : expression2
```

The condition should always evaluate to 1 (true) or 0 (false). If the condition evaluates to true, then **expression1** is evaluated and its value is returned, otherwise **expression2** is evaluated and its value is returned.

Given below is an example demonstrating the usage of a **conditional operator**:

```
int marks = 75, pass_marks = 50;
(marks > pass_marks) ? printf("Passed C Certification") : printf("Failed C Certification");
```

Since the condition marks > pass_marks evaluates to true, the **expression1** containing the **printf** statement in the above example prints "Passed C Certification"

Read the code given below and retype in the space provided.

```
#include <stdio.h>

int main()
{
    int marks = 75, pass_marks = 50;
    (marks > pass_marks) ? printf("Passed C exam.") : printf("Failed C exam.");
    return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int marks=75,pass_marks=50;
    (marks>pass_marks)?printf("Passed C exam."):printf("Failed C exam.");
    return 0;
}
```

	Expected	Got	
✔	Passed C exam.	Passed C exam.	✔

Passed all tests! ✔

In the program given below, fill in the missing code to find the **largest** of the two given numbers using **ternary operator**.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>
```

Started on	Sunday, 1 March 2026, 3:05 PM
State	Finished
Completed on	Sunday, 1 March 2026, 3:12 PM
Time taken	7 mins 7 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

C language provides an operator to evaluate conditions. It is called a **conditional operator** (?:) or a **ternary operator**.

Ternary operator needs exactly **three** operands to compute the result.

The syntax for using a **ternary operator** is:

```
condition? expression1 : expression2
```

The condition should always evaluate to 1 (true) or 0 (false). If the condition evaluates to true, then **expression1** is evaluated and its value is returned, otherwise **expression2** is evaluated and its value is returned.

Given below is an example demonstrating the usage of a **conditional operator**:

```
int marks = 75, pass_marks = 50;
(marks > pass_marks) ? printf("Passed C Certification") : printf("Failed C Certification");
```

Since the condition marks > pass_marks evaluates to true, the **expression1** containing the **printf** statement in the above example prints "Passed C Certification"

Read the code given below and retype in the space provided.

```
#include <stdio.h>

int main()
{
   int marks = 75, pass_marks = 50;
   (marks > pass_marks) ? printf("Passed C exam.") : printf("Failed C exam.");
   return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int marks=75,pass_marks=50;
    (marks>pass_marks)?printf("Passed C exam."):printf("Failed C exam.");
    return 0;
}
```

	Expected	Got	
✔	Passed C exam.	Passed C exam.	✔

Passed all tests! ✔

Question **2**
Correct
Marked out of 1.00
Flag question

In the program given below, fill in the missing code to find the **largest** of the two given numbers using **ternary operator**.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>
```

```
int main()
{
    int marks = 75, pass_marks = 50;
    (marks > pass_marks) ? printf("Passed C exam.") : printf("Failed C exam.");
    return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int marks=75,pass_marks=50;
    (marks>pass_marks)?printf("Passed C exam."):printf("Failed C exam.");
    return 0;
}
```

	Expected	Got	
✔	`Passed C exam.`	`Passed C exam.`	✔

Passed all tests! ✔

In the program given below, fill in the missing code to find the **largest** of the two given numbers using **ternary operator**.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    int num1 = 20, num2 = 25, large;
    large = (num1>num2) ? num1 : num2; // Write the correct code
    printf("Largest number = %d", large);
    return 0;
}
```

	Expected	Got	
✔	`Largest number = 25`	`Largest number = 25`	✔

Passed all tests! ✔

Saved

Finish review

```
int main()
{
    int marks = 75, pass_marks = 50;
    (marks > pass_marks) ? printf("Passed C exam.") : printf("Failed C exam.");
    return 0;
}
```

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int marks=75,pass_marks=50;
    (marks>pass_marks)?printf("Passed C exam."):printf("Failed C exam.");
    return 0;
}
```

	Expected	Got	
✔	Passed C exam.	Passed C exam.	✔

Passed all tests! ✔

Question **2**

Correct

Marked out of 1.00

Flag question

In the program given below, fill in the missing code to find the **largest** of the two given numbers using **ternary operator**.

Answer: (penalty regime: 0 %)

Reset answer

```
#include <stdio.h>

int main()
{
    int num1 = 20, num2 = 25, large;
    large = (num1>num2) ? num1 : num2; // Write the correct code
    printf("Largest number = %d", large);
    return 0;
}
```

	Expected	Got	
✔	Largest number = 25	Largest number = 25	✔

Passed all tests! ✔

Fundamentals of Data Structures Using C-2025 Batcl

- Sunday, 1 March 2026, 3:17 PM
- Finished
- Sunday, 1 March 2026, 3:21 PM
- 4 mins 17 secs
- PAVITHRA B 2025-BME

There was a large ground in center of the city which is rectangular in shape. The Corporation decides to build a Cricket stadium in the area for school and college students, But the area was used as a car parking zone. In order to protect the land from using as an unauthorized parking zone, the corporation wanted to protect the stadium by building a fence. In order to help the workers to build a fence, they planned to place a thick rope around the ground. They wanted to buy only the exact length of the rope that is needed. They also wanted to cover the entire ground with a carpet during rainy season. They wanted to buy only the exact quantity of carpet that is needed. They requested your help. Can you please help them by writing a program to find the exact length of the rope and the exact quantity of carpet that is required?

Input format:

Input consists of 2 integers. The first integer corresponds to the length of the ground and the second integer corresponds to the breadth of the ground.

Output Format:

Output Consists of two integers. The first integer corresponds to the length. The second integer corresponds to the quantity of carpet required.

Sample Input:

50

20

Sample Output:

140

1000

For example:

Input	Result
50 20	140 1000

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
#include <stdio.h>
int main()
{
    int l,b;
    scanf("%d",&l);
    scanf("%d",&b);
    int p=2*(l+b);
    int a=l*b;
    printf("%d\n",p);
    printf("%d",a);
    return 0;
}
```

	Input	Expected	Got	
✔	50 20	140 1000	140 1000	✔

Passed all tests! ✔

Finish review

CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Sunday, 1 March 2026, 3:17 PM
State	Finished
Completed on	Sunday, 1 March 2026, 3:21 PM
Time taken	4 mins 17 secs
Name	PAVITHRA B 2025-BME

Question **1**

Correct

Marked out of 1.00

Flag question

There was a large ground in center of the city which is rectangular in shape. The Corporation decides to build a Cricket stadium in the area for school and college students, But the area was used as a car parking zone. In order to protect the land from using as an unauthorized parking zone, the corporation wanted to protect the stadium by building a fence. In order to help the workers to build a fence, they planned to place a thick rope around the ground. They wanted to buy only the exact length of the rope that is needed. They also wanted to cover the entire ground with a carpet during rainy season. They wanted to buy only the exact quantity of carpet that is needed. They requested your help. Can you please help them by writing a program to find the exact length of the rope and the exact quantity of carpet that is required?

Input format:

Input consists of 2 integers. The first integer corresponds to the length of the ground and the second integer corresponds to the breadth of the ground.

Output Format:

Output Consists of two integers. The first integer corresponds to the length. The second integer corresponds to the quantity of carpet required.

Sample Input:

50

20

Sample Output:

140

1000

For example:

Input	Result
50 20	140 1000

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?
Falling back to raw text area.

```
#include <stdio.h>
int main()
{
    int l,b;
    scanf("%d",&l);
    scanf("%d",&b);
    int p=2*(l+b);
    int a=l*b;
    printf("%d\n",p);
    printf("%d",a);
    return 0;
}
```

	Input	Expected	Got	
✔	50 20	140 1000	140 1000	✔

Passed all tests! ✔

Finish review

Quiz navigation

-undamentals of Data Structures Using C-2025 Batcl

n Sunday, 1 March 2026, 3:24 PM
e Finished
n Sunday, 1 March 2026, 3:27 PM
n 2 mins 24 secs
e PAVITHRA B 2025-BME

Training for sports day has begun and the physical education teacher has decided to conduct some team games. The teacher wants to split the students in higher secondary into equal sized teams. In some cases, there may be some students who are left out from the teams and he wanted to use the left out students to assist him in conducting the team games. For instance, if there are 50 students in a class and if the class has to be divided into 7 equal sized teams, 7 students will be there in each team and 1 student will be left out. That 1 student will assist the PET. With this idea in mind, the PET wants your help to automate this team splitting task. Can you please help him out?

INPUT FORMAT:

Input consists of 2 integers. The first integer corresponds to the number of students in the class and the second integer corresponds to the number of teams.

OUTPUT FORMAT:

The output consists of two integers. The first integer corresponds to the number of students in each team and the second integer corresponds to the students who are left out.

SAMPLE INPUT:

60

8

SAMPLE OUTPUT:

7

4

For example:

Input	Result
60 8	7 4

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main()
{
    int n,t;
    scanf("%d",&n);
    scanf("%d",&t);
    printf("%d\n",n/t);
    printf("%d",n%t);
    return 0;
}
```

	Input	Expected	Got	
✔	60 8	7 4	7 4	✔

Passed all tests! ✔

Finish review

Started on	Sunday, 1 March 2026, 3:24 PM
State	Finished
Completed on	Sunday, 1 March 2026, 3:27 PM
Time taken	2 mins 24 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

Training for sports day has begun and the physical education teacher has decided to conduct some team games. The teacher wants to split the students in higher secondary into equal sized teams. In some cases, there may be some students who are left out from the teams and he wanted to use the left out students to assist him in conducting the team games. For instance, if there are 50 students in a class and if the class has to be divided into 7 equal sized teams, 7 students will be there in each team and 1 student will be left out. That 1 student will assist the PET. With this idea in mind, the PET wants your help to automate this team splitting task. Can you please help him out?

INPUT FORMAT:

Input consists of 2 integers. The first integer corresponds to the number of students in the class and the second integer corresponds to the number of teams.

OUTPUT FORMAT:

The output consists of two integers. The first integer corresponds to the number of students in each team and the second integer corresponds to the students who are left out.

SAMPLE INPUT:

60

8

SAMPLE OUTPUT:

7

4

For example:

Input	Result
60 8	7 4

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int n,t;
    scanf("%d",&n);
    scanf("%d",&t);
    printf("%d\n",n/t);
    printf("%d",n%t);
    return 0;
}
```

	Input	Expected	Got	
✔	60 8	7 4	7 4	✔

Passed all tests! ✔

Finish review

Quiz navigation

1

Fundamentals of Data Structures Using C-2025 Batc

n	Sunday, 1 March 2026, 3:28 PM
e	Finished
n	Sunday, 1 March 2026, 3:32 PM
n	3 mins 35 secs
e	PAVITHRA B 2025-BME

Each Sunday, a newspaper agency sells w copies of a special edition newspaper for Rs.x per copy. The cost to the agency of each newspaper is Rs.y. The agency pays a fixed cost for storage, delivery and so on of Rs.100 per Sunday. The newspaper agency wants to calculate the profit which it obtains only on Sundays. Can you please help them out by writing a program to compute the profit if w, x, and y are given?

INPUT FORMAT:

Input consists of 3 integers: w, x, and y. w is the number of copies sold, x is the cost per copy and y is the cost the agency spends per copy.

OUTPUT FORMAT:

The output consists of a single integer which corresponds to the profit obtained by the newspaper agency.

SAMPLE INPUT:

1000

2

1

SAMPLE OUTPUT:

900

For example:

Input	Result
1000 2 1	900

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int w,x,y;
    scanf("%d%d%d",&w,&x,&y);
    printf("%d",(x*w)-(w*y)-100);
    return 0;
}
```

	Input	Expected	Got	
✔	1000 2 1	900	900	✔

Passed all tests! ✔

Finish review

CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Sunday, 1 March 2026, 3:28 PM
State	Finished
Completed on	Sunday, 1 March 2026, 3:32 PM
Time taken	3 mins 35 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

Each Sunday, a newspaper agency sells w copies of a special edition newspaper for Rs.x per copy. The cost to the agency of each newspaper is Rs.y. The agency pays a fixed cost for storage, delivery and so on of Rs.100 per Sunday. The newspaper agency wants to calculate the profit which it obtains only on Sundays. Can you please help them out by writing a program to compute the profit if w, x, and y are given?

INPUT FORMAT:

Input consists of 3 integers: w, x, and y. w is the number of copies sold, x is the cost per copy and y is the cost the agency spends per copy.

OUTPUT FORMAT:

The output consists of a single integer which corresponds to the profit obtained by the newspaper agency.

SAMPLE INPUT:

1000

2

1

SAMPLE OUTPUT:

900

For example:

Input	Result
1000 2 1	900

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main()
{
    int w,x,y;
    scanf("%d%d%d",&w,&x,&y);
    printf("%d",(x*w)-(w*y)-100);
    return 0;
}
```

	Input	Expected	Got	
✔	1000 2 1	900	900	✔

Passed all tests! ✔

Finish review

Quiz navigation

1

Finish review

-undamentals of Data Structures Using C-2025 Batcl

ı Sunday, 1 March 2026, 3:35 PM
e Finished
ı Sunday, 1 March 2026, 3:36 PM
ı 1 min 35 secs
e PAVITHRA B 2025-BME

Four kids Peter, Susan, Edmond and Lucy travel through a wardrobe to the land of Narnia. Narnia is a fantasy world of magic with mythical beasts and talking animals.While exploring the land of narnia Lucy found Mr.Tumnus the two legged stag ,and she followed it, down a narrow path .She and Mr.Tumnus became friends and he offered a cup of coffee to Lucy in his small hut.It was time for Lucy to return to her family and so she bid good bye to Mr.Tumnus and while leaving Mr.Tumnus told that it is quite difficult to find the route back as it was already dark. He told her to see the trees while returning back and said that the first tree with two digits number will help her find the way and the way to go back to her home is the sum of digits of the tree and that numbered way will lead her to the tree next to the wardrobe where she can find the others. Lucy was already confused, so please help her in finding the route to her home....

Input Format:

Input consists of an integer corresponding to the 2-digit number.

Output Format:

Output consists of an integer corresponding to the sum of its digits.

SAMPLE INPUT:

87

SAMPLE OUTPUT:

15

For example:

Input	Result
87	15

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    printf("%d",(n/10)+(n%10));
    return 0;
}
```

	Input	Expected	Got	
✔	87	15	15	✔
✔	54	9	9	✔

Passed all tests! ✔

Finish review

CS23232-Fundamentals of Data Structures Using C-2025 Batch

Started on	Sunday, 1 March 2026, 3:35 PM
State	Finished
Completed on	Sunday, 1 March 2026, 3:36 PM
Time taken	1 min 35 secs
Name	PAVITHRA B 2025-BME

Question **1**
Correct
Marked out of 1.00
Flag question

Four kids Peter, Susan, Edmond and Lucy travel through a wardrobe to the land of Narnia. Narnia is a fantasy world of magic with mythical beasts and talking animals.While exploring the land of narnia Lucy found Mr.Tumnus the two legged stag ,and she followed it, down a narrow path .She and Mr.Tumnus became friends and he offered a cup of coffee to Lucy in his small hut.It was time for Lucy to return to her family and so she bid good bye to Mr.Tumnus and while leaving Mr.Tumnus told that it is quite difficult to find the route back as it was already dark. He told her to see the trees while returning back and said that the first tree with two digits number will help her find the way and the way to go back to her home is the sum of digits of the tree and that numbered way will lead her to the tree next to the wardrobe where she can find the others. Lucy was already confused, so please help her in finding the route to her home....

Input Format:

Input consists of an integer corresponding to the 2-digit number.

Output Format:

Output consists of an integer corresponding to the sum of its digits.

SAMPLE INPUT:

87

SAMPLE OUTPUT:

15

For example:

Input	Result
87	15

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    printf("%d",(n/10)+(n%10));
    return 0;
}
```

	Input	Expected	Got	
✔	87	15	15	✔
✔	54	9	9	✔

Passed all tests! ✔

Finish review

Quiz navigation

1

Finish review